

Optimization: Critical Points & Lagrange Multipliers

ANSWER KEY



Finding critical points

- Find the critical point(s) of $f(x, y) = x^2 + 3y^2 - 4x + 6y$.
 $f_x = 2x - 4 = 0 \Rightarrow x = 2$. $f_y = 6y + 6 = 0 \Rightarrow y = -1$. Critical point: $(2, -1)$.
- Find the critical point(s) of $f(x, y) = xy - x^2 - y^2$.
 $f_x = y - 2x = 0 \Rightarrow y = 2x$. $f_y = x - 2y = 0 \Rightarrow x = 2y$. Substituting: $x = 2(2x) = 4x \Rightarrow x = 0$, so $y = 0$.
Critical point: $(0, 0)$.
- Find the critical point(s) of $f(x, y) = x^3 - 3xy + y^3$.
 $f_x = 3x^2 - 3y = 0 \Rightarrow y = x^2$. $f_y = -3x + 3y^2 = 0 \Rightarrow x = y^2$. Substituting $y = x^2$ into $x = y^2$:
 $x = x^4 \Rightarrow x^4 - x = 0 \Rightarrow x(x^3 - 1) = 0 \Rightarrow x = 0$ or $x = 1$. Critical points: $(0, 0)$ and $(1, 1)$.

The Hessian and the Second Derivative Test

- Classify the critical point $(2, -1)$ of $f(x, y) = x^2 + 3y^2 - 4x + 6y$ using $D = f_{xx}f_{yy} - (f_{xy})^2$.
 $f_{xx} = 2$, $f_{yy} = 6$, $f_{xy} = 0$. $D = 2(6) - 0 = 12 > 0$ and $f_{xx} > 0$, so $(2, -1)$ is a local minimum.
- Classify the critical point $(0, 0)$ of $f(x, y) = xy - x^2 - y^2$.
 $f_{xx} = -2$, $f_{yy} = -2$, $f_{xy} = 1$. $D = (-2)(-2) - 1^2 = 3 > 0$ and $f_{xx} < 0$, so $(0, 0)$ is a local maximum.
- Classify the critical point of $f(x, y) = x^2 - y^2$.
 $f_x = 2x = 0$, $f_y = -2y = 0 \Rightarrow (0, 0)$. $f_{xx} = 2$, $f_{yy} = -2$, $f_{xy} = 0$. $D = 2(-2) - 0 = -4 < 0$, so $(0, 0)$ is a saddle point.
- Classify both critical points $(0, 0)$ and $(1, 1)$ of $f(x, y) = x^3 - 3xy + y^3$ (found above).
 $f_{xx} = 6x$, $f_{yy} = 6y$, $f_{xy} = -3$. At $(0, 0)$: $D = 0(0) - 9 = -9 < 0$, a saddle point. At $(1, 1)$:
 $D = 6(6) - 9 = 27 > 0$ and $f_{xx} = 6 > 0$, a local minimum.

When the Second Derivative Test fails

- Show that the Second Derivative Test is inconclusive at the critical point $(0, 0)$ of $f(x, y) = x^4 + y^4$.
 $f_x = 4x^3$, $f_y = 4y^3$, both zero at $(0, 0)$. $f_{xx} = 12x^2$, $f_{yy} = 12y^2$, $f_{xy} = 0$, all zero at $(0, 0)$, so $D = 0$ — the test gives no information.
- Determine directly (without the Second Derivative Test) whether $(0, 0)$ is a local min, max, or saddle for $f(x, y) = x^4 + y^4$.
Since $x^4 \geq 0$ and $y^4 \geq 0$ for all (x, y) , $f(x, y) \geq 0 = f(0, 0)$ everywhere, so $(0, 0)$ is a (global, hence local) minimum.

Lagrange multipliers: setting up the system

- 10** Set up the Lagrange multiplier equations to optimize $f(x, y) = xy$ subject to $g(x, y) = x + y - 10 = 0$.
 $\nabla f = \lambda \nabla g: \langle y, x \rangle = \lambda \langle 1, 1 \rangle \Rightarrow y = \lambda, x = \lambda \Rightarrow x = y$, together with the constraint $x + y = 10$.
- 11** Set up the Lagrange multiplier equations to optimize $f(x, y) = x^2 + y^2$ subject to $g(x, y) = xy - 1 = 0$.
 $\nabla f = \lambda \nabla g: \langle 2x, 2y \rangle = \lambda \langle y, x \rangle \Rightarrow 2x = \lambda y$ and $2y = \lambda x$, together with the constraint $xy = 1$.
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Solving Lagrange multiplier problems

- 12** Maximize $f(x, y) = xy$ subject to $x + y = 10$.
From the setup above, $x = y$ and $x + y = 10 \Rightarrow x = y = 5$. Maximum value: $f(5, 5) = 25$.
- 13** Minimize $f(x, y) = x^2 + y^2$ subject to $x + y = 4$.
 $\nabla f = \lambda \nabla g: \langle 2x, 2y \rangle = \lambda \langle 1, 1 \rangle \Rightarrow x = y$. With $x + y = 4$: $x = y = 2$. Minimum value: $f(2, 2) = 8$.
- 14** Solve the system from the earlier setup question to minimize $f(x, y) = x^2 + y^2$ subject to $xy = 1$.
From $2x = \lambda y$ and $2y = \lambda x$: multiplying, $4xy = \lambda^2 xy$, and since $xy = 1 \neq 0$, $\lambda^2 = 4 \Rightarrow \lambda = \pm 2$. If $\lambda = 2$: $2x = 2y \Rightarrow x = y$; with $xy = 1$: $x = y = 1$ or $x = y = -1$. Both give $f = 1 + 1 = 2$. (If $\lambda = -2$: $x = -y$, but then $xy = -x^2 < 0 \neq 1$, no real solution.) Minimum value: $f = 2$, at $(1, 1)$ and $(-1, -1)$.
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A geometric Lagrange multiplier application

- 15** Find the point on the line $x + y = 6$ closest to the origin, using Lagrange multipliers on $f(x, y) = x^2 + y^2$.
 $\nabla f = \lambda \nabla g: \langle 2x, 2y \rangle = \lambda \langle 1, 1 \rangle \Rightarrow x = y$. With $x + y = 6$: $x = y = 3$. Closest point: $(3, 3)$, at distance $\sqrt{9 + 9} = 3\sqrt{2}$.
- 16** A rectangle has perimeter 20. Use Lagrange multipliers to maximize its area $A = xy$ subject to the constraint $2x + 2y = 20$ (i.e., $x + y = 10$).
This matches the very first Lagrange problem above: $x = y = 5$, giving maximum area $A = 25$ — a square, as expected.
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Lagrange multipliers with three variables

- 17** Set up (but do not fully solve) the Lagrange multiplier equations to minimize $f(x, y, z) = x^2 + y^2 + z^2$ subject to $g(x, y, z) = x + y + z - 6 = 0$.
 $\nabla f = \lambda \nabla g: \langle 2x, 2y, 2z \rangle = \lambda \langle 1, 1, 1 \rangle \Rightarrow 2x = 2y = 2z = \lambda$, so $x = y = z$, together with $x + y + z = 6$.
- 18** Solve the three-variable problem set up above.
From $x = y = z$ and $x + y + z = 6$: $3x = 6 \Rightarrow x = y = z = 2$. Minimum value: $f(2, 2, 2) = 4 + 4 + 4 = 12$.

Interpreting the meaning of λ

- 19** In the rectangle-area problem above ($x = y = 5$, $\lambda = 5$), briefly explain what λ represents in economic/physical terms: the approximate change in the optimal value of f per unit increase in the constraint constant.
- λ approximates $\frac{dA^*}{dc}$, where A^* is the optimal (maximum) area and c is the perimeter constant (here $c = 20$, giving $x + y = 10$). So increasing the perimeter by 1 unit (from 20 to 21) would increase the maximum enclosed area by approximately $\lambda = 5$.
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Optimization on a closed region (global extrema)

- 20** Explain the general procedure for finding the global maximum and minimum of a continuous $f(x, y)$ on a closed, bounded region R .
- (1) Find all critical points of f in the interior of R and evaluate f there. (2) Find the extrema of f restricted to the boundary of R (often via substitution or Lagrange multipliers) and evaluate f at those boundary extrema, including corners. (3) Compare all these values — the largest is the global maximum, the smallest is the global minimum (guaranteed to exist by the Extreme Value Theorem for continuous f on a closed, bounded region).
- 21** Find the only interior critical point of $f(x, y) = x^2 + y^2 - 2x$ on the closed disk $x^2 + y^2 \leq 4$, and evaluate f there.
- $f_x = 2x - 2 = 0 \Rightarrow x = 1$. $f_y = 2y = 0 \Rightarrow y = 0$. Critical point $(1, 0)$, which lies inside the disk since $1^2 + 0^2 = 1 < 4$. $f(1, 0) = 1 + 0 - 2 = -1$.
- 22** For the same $f(x, y) = x^2 + y^2 - 2x$ on $x^2 + y^2 \leq 4$, find the maximum of f on the boundary circle $x^2 + y^2 = 4$ using Lagrange multipliers.
- $\nabla f = \lambda \nabla g$ with $g = x^2 + y^2 - 4$: $\langle 2x - 2, 2y \rangle = \lambda \langle 2x, 2y \rangle$. From the y -component: $2y = 2\lambda y \Rightarrow y = 0$ or $\lambda = 1$. If $y = 0$: $x^2 = 4 \Rightarrow x = \pm 2$, giving $f(2, 0) = 4 - 4 = 0$ and $f(-2, 0) = 4 + 4 = 8$. If $\lambda = 1$: $2x - 2 = 2x \Rightarrow -2 = 0$, impossible. So on the boundary, the maximum is $f(-2, 0) = 8$.
- 23** Combine the results above to state the global maximum and minimum of $f(x, y) = x^2 + y^2 - 2x$ on the disk $x^2 + y^2 \leq 4$.
- Candidates: interior critical point $f(1, 0) = -1$; boundary values $f(2, 0) = 0$ and $f(-2, 0) = 8$. Global minimum: -1 at $(1, 0)$. Global maximum: 8 at $(-2, 0)$.
- 24** Explain why the Extreme Value Theorem guarantees this problem has a global max and min, and why that guarantee would fail on the OPEN disk $x^2 + y^2 < 4$.
- The Extreme Value Theorem requires the region to be closed and bounded; the closed disk $x^2 + y^2 \leq 4$ satisfies both. The open disk $x^2 + y^2 < 4$ is bounded but not closed (it excludes its boundary circle), so a continuous function need not attain a maximum or minimum on it — e.g., f could approach its boundary supremum without ever reaching it inside the open region.

- 25** Find the global maximum and minimum of $f(x, y) = xy$ on the closed square $-1 \leq x \leq 1$, $-1 \leq y \leq 1$, by checking the interior critical point and all four corners (the boundary edges of a square are easiest checked at their endpoints for a bilinear function like xy).

Interior: $f_x = y = 0$, $f_y = x = 0 \Rightarrow (0, 0)$, with $f(0, 0) = 0$. Corners: $f(1, 1) = 1$, $f(1, -1) = -1$, $f(-1, 1) = -1$, $f(-1, -1) = 1$. Global maximum: 1 (at $(1, 1)$ and $(-1, -1)$). Global minimum: -1 (at $(1, -1)$ and $(-1, 1)$).